

# Pulkit Verma

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## Research Interests

AI Safety, AI Planning, Interpretability, Symbolic Model Learning, Analysis of Abstractions, and Robotics.

## Education

|                                                  |                                                                                                                                                                                                                                   |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ph.D.</b><br>Computer Science<br>2018-2024    | <b>Arizona State University</b> , School of Computing and Augmented Intelligence<br>Thesis: <i>Data-Efficient Paradigms for Personalized Assessment of Taskable AI Systems</i> <a href="#">↗</a><br>Advisor: Siddharth Srivastava |
| <b>M.Tech.</b><br>Computer Science<br>2013-2015  | <b>Indian Institute of Technology Guwahati</b> , Department of CSE<br>Thesis: <i>Resource Usage Analysis for Speech Recognition Techniques</i> <a href="#">↗</a><br>Advisor: Pradip K. Das                                        |
| <b>B.Engg.</b><br>Information Tech.<br>2007-2011 | <b>Shri Vaishnav Institute of Technology &amp; Science, Indore</b> , Department of IT<br>Capstone: Smart Analyzer - Predict and Cache Webpages based on Usage<br>Advisor: Anand S. Rajawat                                        |

## Research and Professional Experience

|                      |                                                                                                                                                                                                    |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sep 2024 – (present) | <b>Postdoctoral Research Associate</b> , CSAIL, Massachusetts Institute of Technology.<br>Advisor: Julie Shah.<br><i>Research Areas:</i> AI Interpretability, Model Learning, Black-Box Assessment |
| Aug 2018 – Aug 2024  | <b>Graduate Research Associate</b> , AAIR Lab, ASU.<br>Advisor: Siddharth Srivastava.<br><i>Research Areas:</i> Agent Assessment, Model Learning, Automated Planning                               |
| May 2023 – Aug 2023  | <b>Research Scientist Intern</b> , Meta AI.<br>Mentors: Rohan Chitnis, Alborz Geramifard, Nitin Kamra.<br><i>Research Areas:</i> Automated Planning, Curriculum Personalisation                    |
| Jul 2015 – Jul 2018  | <b>Member of Technical Staff</b> , NetApp Inc.<br><i>Predicted workloads</i> for storage volumes using I/O patterns for All Flash systems.                                                         |
| Aug 2013 – May 2015  | <b>Research Assistant</b> , IIT Guwahati.<br>Mentors: Pradip K. Das, Shivashankar B. Nair, Shashi Shekhar Jha.<br><i>Research Areas:</i> Speech Processing, Mobile Robotics                        |
| Mar 2012 – Aug 2013  | <b>Assistant Systems Engineer</b> , Tata Consultancy Services Pvt. Ltd.<br><i>Developed iOS applications</i> for various B2C companies.                                                            |

## Teaching Experience

|                                                       |                                                                                                                                                                                                           |                                                      |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| <b>Teaching Assistant</b><br>Arizona State University | CSE 571 Artificial Intelligence<br>CSE 571 Artificial Intelligence                                                                                                                                        | Spring 2022<br>Fall 2019                             |
| <b>Session Lead</b><br>Arizona State University       | CSE 571 Artificial Intelligence<br>CSE 471 Introduction to Artificial Intelligence<br>CSE 574 Planning and Learning Methods in Artificial Intelligence<br>CSE 471 Introduction to Artificial Intelligence | Spring 2023<br>Fall 2022<br>Spring 2021<br>Fall 2020 |

## Teaching Experience (continued)

|                                    |                                 |             |
|------------------------------------|---------------------------------|-------------|
| Teaching Assistant<br>IIT Guwahati | CS 243 Software Engineering Lab | Spring 2015 |
|                                    | CS 566 Speech Processing        | Fall 2014   |
|                                    | CS 462 Computer Graphics Lab    | Spring 2014 |
|                                    | CS 513 Programming Lab          | Fall 2013   |

## Honors and Awards

|                                                                                                                                                                                       |               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Graduate College Completion Fellowship</b> , ASU Graduate College <a href="#">↗</a><br>Highly competitive (1 per program per year) with tuition, stipend, and insurance for 1 year | June 2023     |
| <b>Best Demo Award</b> , AAMAS 2022 <a href="#">↗</a><br>Awarded to one system demonstration at the AAMAS 2022 conference                                                             | May 2022      |
| <b>Outstanding Reviewer</b> , CoRL 2024 LEAP Workshop <a href="#">↗</a>                                                                                                               | November 2024 |
| <b>Spring 2024 GPSA Jumpstart Research Grant</b> , ASU GPSA <a href="#">↗</a>                                                                                                         | January 2024  |
| <b>Fall 2023 GPSA Outstanding Research Award</b> , ASU GPSA <a href="#">↗</a>                                                                                                         | October 2023  |
| <b>Fall 2023 GPSA Outstanding Mentor Award</b> , ASU GPSA <a href="#">↗</a>                                                                                                           | October 2023  |
| <b>University (Engineering) Graduate Fellowship</b> , ASU Engineering and Graduate College <a href="#">↗</a>                                                                          | March 2023    |
| <b>SCAI Doctoral Fellowship</b> , School of Computing and Augmented Intelligence, ASU                                                                                                 | March 2023    |
| <b>Spring 2022 GPSA Teaching Excellence Award</b> , ASU GPSA <a href="#">↗</a>                                                                                                        | April 2022    |
| <b>SCAI Doctoral Fellowship</b> , School of Computing and Augmented Intelligence, ASU                                                                                                 | March 2022    |
| <b>Fulton Schools Experiential Learning Grant</b> , ASU Engineering <a href="#">↗</a>                                                                                                 | October 2018  |
| <b>Post Graduation Fellowship</b> , All India Council for Technical Education, Govt. of India <a href="#">↗</a>                                                                       | 2013-2015     |
| <b>TCS ILP Kudos Award</b> , Tata Consultancy Services Ltd. (Best Trainee in the Batch)                                                                                               | May 2012      |

## Talks

|                                                                                                                                                                                                      |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Learning Interpretable Models for Personalized Assessment of Taskable AI Systems</b><br>Research Talk Series, School of Artificial Intelligence and Data Engineering, IIT Ropar <a href="#">↗</a> | October 2024  |
| <b>Learning Interpretable Models for Autonomous Assessment of Taskable AI Systems</b><br>Brown Robotics Talks, BigAI, Brown University <a href="#">↗</a>                                             | October 2024  |
| <b>Learning Interpretable Models for Personalized Assessment of Taskable AI Systems</b><br>Research Seminar, Department of CSE, IIT Guwahati <a href="#">↗</a>                                       | August 2024   |
| <b>Learning Interpretable Models for Personalized Assessment of Taskable AI Systems</b><br>Interactive Robotics Group, Massachusetts Institute of Technology <a href="#">↗</a>                       | May 2024      |
| <b>Learning Symbolic Models for Personalized Assessment of Taskable AI</b><br>ICAROS Lab, University of Southern California <a href="#">↗</a>                                                        | May 2024      |
| <b>Data-Efficient Paradigms for Personalized Assessment of Taskable AI Systems</b><br>CHAI Hands-On, University of California Berkeley <a href="#">↗</a>                                             | April 2024    |
| <b>Data-Efficient Paradigms for Personalized Assessment of Black-Box AI Systems</b><br>Intelligent Robot Learning Lab, University of Alberta <a href="#">↗</a>                                       | February 2024 |
| <b>Discovering User-Interpretable Capabilities of Black-Box AI Agents</b><br>CHAI - 6th Annual Workshop <a href="#">↗</a>                                                                            | June 2022     |

## Publications

\* denotes equal contribution

### Conference and Journal Papers (Archival)

- [C1] Rushang Karia\*, **Pulkit Verma\***, Alberto Speranzon, and Siddharth Srivastava. “Epistemic Exploration for Generalizable Planning and Learning in Non-Stationary Settings”. In *34th International Conference on Automated Planning and Scheduling*, 2024. 
- [C2] **Pulkit Verma**, Rushang Karia, and Siddharth Srivastava. “Autonomous Capability Assessment of Sequential Decision-Making Systems in Stochastic Settings”. In *37th Conference on Advances in Neural Information Processing Systems*, 2023. 
- [C3] **Pulkit Verma**, Shashank Rao Marpally, and Siddharth Srivastava. “Discovering User-Interpretable Capabilities of Black-Box Planning Agents”. In *19th International Conference on Principles of Knowledge Representation and Reasoning*, 2022. 
- [C4] Naman Shah\*, **Pulkit Verma\***, Trevor Angle, and Siddharth Srivastava. “JEDAI: A System for Skill-Aligned Explainable Robot Planning”. In *21st International Conference on Autonomous Agents and MultiAgent Systems (Demonstration Track)*, 2022. **[Best Demo Award]**. 
- [C5] Rashmeet Kaur Nayyar\*, **Pulkit Verma\***, and Siddharth Srivastava. “Differential Assessment of Black-Box AI Agents”. In *36th AAAI Conference on Artificial Intelligence*, 2022. 
- [C6] Yizhong Wang, Swaroop Mishra, Pegah Alipoormolabashi, Yeganeh Kordi, Amirreza Mirzaei, Anjana Arunkumar, Arjun Ashok, **Pulkit Verma**, et al. “Super-NaturalInstructions: Generalization via Declarative Instructions on 1600+ Tasks”. In *2022 Conference on Empirical Methods in Natural Language Processing*, 2022. 
- [C7] **Pulkit Verma**, Shashank R Marpally, and Siddharth Srivastava. “Asking the Right Questions: Learning Interpretable Action Models Through Query Answering”. In *35th AAAI Conference on Artificial Intelligence*, 2021. 
- [C8] **Pulkit Verma**, and Pradip K. Das. “A Comparative Study of Resource Usage for Speaker Recognition Techniques”. In *International Conference on Signal Processing and Communication*, 2016. 
- [C9] **Pulkit Verma**, and Pradip K. Das. “i-Vectors in Speech Processing Applications: A Survey”. In *International Journal of Speech Technology*, 18(4), pp.529-546, 2015. 
- [C10] Mayank Gupta, **Pulkit Verma**, Tuhin Bhattacharya, and Pradip K. Das. “A Mobile Agents based Distributed Speech Recognition Engine for Controlling Multiple Robots”. In *International Conference on Advances In Robotics*, 2015. 
- [C11] **Pulkit Verma**, Mayank Gupta, Tuhin Bhattacharya, and Pradip K. Das. “Improving Services using Mobile Agents-based IoT in a Smart City”. In *International Conference on Contemporary Computing and Informatics*, 2014. 

### Peer-Reviewed Workshop and Symposia Papers (Non-archival)

- [W1] **Pulkit Verma** and Siddharth Srivastava. “User-Aligned Autonomous Capability Assessment of Black-Box AI Systems”. In *AAAI Spring Symposium on User-Aligned Assessment of Adaptive AI Systems*, 2024. 
- [W2] Rushang Karia, Daksh Dobhal, Daniel Bramblett, **Pulkit Verma**, and Siddharth Srivastava. “Can LLMs translate SATisfactorily? Assessing LLM Capabilities in Generating Formal Specifications”. In *AAAI Spring Symposium on User-Aligned Assessment of Adaptive AI Systems*, 2024. 
- [W3] **Pulkit Verma\***, Rushang Karia\*, Gaurav Vipat, Anmol Gupta, and Siddharth Srivastava. “Learning AI-System Capabilities under Stochasticity”. In *NeurIPS Workshop on Generalization in Planning*, 2023. 
- [W4] Rushang Karia\*, **Pulkit Verma\***, Gaurav Vipat, and Siddharth Srivastava. “Epistemically Guided Exploration for Transferable Stochastic Planning in Non-stationary Relational Settings”. In *NeurIPS Workshop on Generalization in Planning*, 2023. 
- [W5] **Pulkit Verma** and Siddharth Srivastava. “Learning Causal Models of Autonomous Agents using Interventions”. In *IJCAI Workshop on Generalization in Planning*, 2021. 

- [W6] **Pulkit Verma**, Shashank Rao Marpally, and Siddharth Srivastava. “Learning User-Interpretable Descriptions of Black-Box AI System Capabilities”. In *ICAPS Workshop on Knowledge Engineering for Planning & Scheduling*, 2021. 
- [W7] **Pulkit Verma** and Siddharth Srivastava. “Learning Interpretable Models for Black-Box Agents”. In *ICML Workshop on Human In the Loop Learning*, 2020. 
- [W8] **Pulkit Verma**, and Siddharth Srivastava. “Learning Generalized Models by Interrogating Black-Box Autonomous Agents”. In *AAAI Workshop on Generalization in Planning*, 2020. 
- [W9] Alok Shankar Mysore, Vikas S. Yaligar, Imanol Arrieta Ibarra, Camelia Simoiu, Sharad Goel, Ramesh Arvind, Chiraag Sumanth, Arvind Srikantan, Bhargav HS, Mayank Pahadia, Tushar Dobha, Atif Ahmed, Mani Shankar, **Pulkit Verma**, et al. “Investigating the ‘Wisdom of Crowds’ at Scale.”. In *Adjunct Proceedings of the 28th Annual ACM Symposium on User Interface Software and Technology (Poster)*, 2015. 

## Preprints/Under Review

- [P1] Rushang Karia, Daniel Bramblett, Daksh Dobhal, **Pulkit Verma**, and Siddharth Srivastava. “ $\forall$ uto $\exists$ val: Autonomous Assessment of LLMs in Formal Synthesis and Interpretation Tasks”. 2024. 
- [P2] Naman Shah, Jayesh Nagpal, **Pulkit Verma**, and Siddharth Srivastava. “From Reals to Logic and Back: Inventing Symbolic Vocabularies, Actions, and Models for Planning from Raw Data”. 2024. 
- [P3] Mehdi Dadvar, Keyvan Majd, Elena Oikonomou, **Pulkit Verma**, Georgios Fainekos, and Siddharth Srivastava. “Joint Communication and Motion Planning for Cobots in Real-World Contexts”. 2024.
- [P4] Daksh Dobhal, Jayesh Nagpal, Rushang Karia, **Pulkit Verma**, Rashmeet K Nayyar, Naman Shah, and Siddharth Srivastava. “Using Explainable AI and Hierarchical Planning for Outreach with Robots”. 2024. 

## Doctoral Consortium

- [D1] **Pulkit Verma**. “Data Efficient Paradigms for Personalized Assessment of Black-Box Taskable AI Systems”. In *38th AAAI Conference on Artificial Intelligence*, 2024.   
Mentor: Sheila McIlraith
- [D2] **Pulkit Verma**. “Sample Efficient Paradigms for Personalized Assessment of Taskable AI Systems”. In *32nd International Joint Conference on Artificial Intelligence*, 2023.   
Mentor: Peter Stone
- [D3] **Pulkit Verma**. “Data Efficient Paradigms for Personalized Assessment of Taskable AI Systems – Dissertation Abstract”. In *32nd International Conference on Automated Planning and Scheduling*, 2022.   
Mentor: Eva Onaindia
- [D4] **Pulkit Verma**. “Data Efficient Paradigms for Personalized Assessment of Taskable AI Systems”. In *19th International Conference on Principles of Knowledge Representation and Reasoning*, 2022.  
Mentor: Hector Geffner
- [D5] **Pulkit Verma**. “Data Efficient Algorithms and Interpretability Requirements for Personalized Assessment of Taskable AI Systems”. In *30th International Joint Conference on Artificial Intelligence*, 2021.  
Mentor: Leslie Pack Kaelbling

## Technical Reports

- [TR1] **Pulkit Verma** and Siddharth Srivastava. “Learning Causally Accurate Models for Autonomous Assessment of Deterministic Black-Box Agents”. Technical Report TR-ASUSCAI-2024-001, School of Computing and Augmented Intelligence, Arizona State University, 2024. 

## Theses

- [T1] **Pulkit Verma**. “Data-Efficient Paradigms for Personalized Assessment of Taskable AI Systems”. (PhD Dissertation). Arizona State University, Tempe, AZ. 
- [T2] **Pulkit Verma**. “Resource Usage Analysis for Speech Recognition Techniques”. (Masters Thesis). Indian Institute of Technology Guwahati, Guwahati, India. 

## Intellectual Property

[IP1] Siddharth Srivastava and **Pulkit Verma**. “Systems and Methods for Independent Audit and Assessment Framework for AI Systems”. US Patent Application 63/597,259.

## Academic Service

### Conference Tutorials

**AAAI 2025 Tutorial on User-Driven Capability Assessment of Taskable AI Systems** [↗](#) 2025  
Pulkit Verma and Siddharth Srivastava

### Workshop Leadership

**AAAI 2024 Spring Symposium on User-Aligned Assessment of Adaptive AI Systems** [↗](#) 2023-2024  
Co-organizers: Rohan Chitnis, Georgios Fainekos, Hazem Torfah, Siddharth Srivastava

**NeurIPS 2023 Workshop on Generalization in Planning (GenPlan’23)** [↗](#) 2023  
Co-organizers: Siddharth Srivastava, Aviv Tamar, Felipe W. Trevizan

**IJCAI 2022 Workshop on Generalization in Planning (GenPlan’22)** [↗](#) 2022  
Co-organizers: Yuqian Jiang, Rushang Karia, Jendrik Seipp

### Journal Reviewer

Neural Computing and Applications (NCAA) [↗](#) 2024-(present)  
Journal of Artificial Intelligence Research (JAIR) [↗](#) 2023-(present)  
IEEE Robotics and Automation Letters (RA-L) [↗](#) 2021-(present)

### Conference Program Committee/Reviewer

AAAI Conference on Artificial Intelligence (AAAI) 2023 [↗](#), 2024 [↗](#), 2025 [↗](#)  
International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS) 2023 [↗](#), 2024 [↗](#), 2025 [↗](#)  
International Conference on Automated Planning and Scheduling (ICAPS) 2022 [↗](#), 2024 [↗](#), 2025 [↗](#)  
International Conference on Learning Representations (ICLR) 2024 [↗](#), 2025 [↗](#)  
Advances in Neural Information Processing System (NeurIPS) 2023 [↗](#), 2024 [↗](#)  
International Joint Conference on Artificial Intelligence (IJCAI) 2023 [↗](#), 2024 [↗](#)  
International Conference on Artificial Intelligence and Statistics (AISTATS) 2025 [↗](#)  
Robotics: Science and Systems Conference (R:SS) 2024 [↗](#)

### Workshop Program Committee/Reviewer

IJCAI/AAAI Workshop on Generalization in Planning (GenPlan) 2021 [↗](#), 2025 [↗](#)  
AAAI/IJCAI Workshop on Bridging the Gap Between AI Planning and RL (PRL) 2023 [↗](#), 2025 [↗](#)  
CoRL Workshop on Learning Effective Abstractions for Planning (LEAP) 2023 [↗](#), 2024 [↗](#)  
NeurIPS Workshop on Explainable/Interpretable AI in Action (AIA) 2023 [↗](#), 2024 [↗](#)  
ICAPS Workshop on Human-Aware and Explainable Planning (HAXP) 2023 [↗](#)  
CVPR Workshop on Open-Domain Reasoning Under Multi-Modal Settings (ODRUM) 2023 [↗](#)  
ICAPS Workshop on Explainable AI Planning (XAIP) 2021 [↗](#), 2022 [↗](#)

## Academic Service (continued)

### Miscellaneous Academic Service

|                                                                                     |           |
|-------------------------------------------------------------------------------------|-----------|
| Co-author, ONR grant on Adaptive Training and Education for Adaptive AI Systems     | 2023      |
| Member, (Inaugural) PhD Council, School of Computing and AI, ASU                    | 2023-2024 |
| Award and Grant Reviewer, Graduate and Professional Service Association (GPSA), ASU | 2020-2022 |

### Volunteer

|                                                                                     |                                                 |
|-------------------------------------------------------------------------------------|-------------------------------------------------|
| International Joint Conference on Artificial Intelligence (IJCAI)                   | 2021 <a href="#">↗</a> , 2022 <a href="#">↗</a> |
| International Conference on Principles of Knowledge Representation & Reasoning (KR) | 2018 <a href="#">↗</a> , 2022 <a href="#">↗</a> |
| International Conference on Learning Representations (ICLR)                         | 2021 <a href="#">↗</a>                          |
| International Conference on Automated Planning and Scheduling (ICAPS)               | 2021 <a href="#">↗</a>                          |
| Workshop on GPU Programming and Applications, IITG                                  | 2014 <a href="#">↗</a>                          |

## Mentoring

### PhD Mentees

|                  |                                       |                |
|------------------|---------------------------------------|----------------|
| Josh Rountree    | Massachusetts Institute of Technology | 2024-(present) |
| Ngoc La          | Massachusetts Institute of Technology | 2024-(present) |
| Daniel Bramblett | Arizona State University              | 2024-(present) |

### Masters Mentees

|                       |                                       |                |
|-----------------------|---------------------------------------|----------------|
| Samir Wadhwanja       | Massachusetts Institute of Technology | 2024-(present) |
| Jayesh Nagpal         | Arizona State University              | 2022-2024      |
| Daksh Dobhal          | Arizona State University              | 2023-2024      |
| Gaurav Vipat          | Arizona State University              | 2023-2024      |
| Trevor Angle          | Arizona State University              | 2021-2022      |
| Shashank Rao Marpally | Arizona State University              | 2020-2021      |

### Undergrad Mentees

|                  |                                         |           |
|------------------|-----------------------------------------|-----------|
| Dylan Fulop      | Arizona State University                | 2022-2023 |
| Kyle J. Atkinson | Arizona State University                | 2021-2023 |
| Masih Tamsoy     | Indian Institute of Technology Guwahati | 2014-2015 |

## Outreach Activities

|                                                                              |           |
|------------------------------------------------------------------------------|-----------|
| Doctoral Mentor, School of Computing and Augmented Intelligence, ASU         | 2023-2024 |
| Mentored junior PhD students and helped them navigate through various issues |           |
| Volunteer, Child Rights and You <a href="#">↗</a>                            | 2016-2018 |
| Taught and mentored primary school children in Koramangala PAG, Bengaluru    |           |
| Student Volunteer, Stanford Scholar Initiative <a href="#">↗</a>             | 2016-2018 |
| Translated and recorded Python tutorial and research talks in Hindi          |           |

## Other Affiliations

|                                                                                                     |                |
|-----------------------------------------------------------------------------------------------------|----------------|
| <b>Member</b> , Association for the Advancement of Artificial Intelligence (AAAI) <a href="#">↗</a> | 2020-(present) |
| <b>Affiliated Graduate Student</b> , Center for Human-Compatible AI, UCB <a href="#">↗</a>          | 2020-2024      |
| <b>Participant</b> , 3rd Summer School on Cognitive Robotics, USC <a href="#">↗</a>                 | July 2019      |

References Available on Request